Analysis of k-Nearest Neighbors on the Digits Dataset

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1 Introduction

This document presents the implementation and analysis of the k-Nearest Neighbors (k-NN) classifier on the Digits dataset, exploring how varying the number of neighbors (k) affects the model's accuracy.

2 Implementation

The implementation involves loading the Digits dataset, splitting it into training and testing sets with stratification, and then applying the k-NN classifier with different values of k.

3 Results

The classifiers were evaluated based on their accuracy scores. The obtained results are as follows:

- Nearest Neighbor Accuracy: 98.6%
- k-Nearest Neighbors Accuracy for k=3: 98.6%
- k-Nearest Neighbors Accuracy for k=5: 98.6%
- k-Nearest Neighbors Accuracy for k=7: 98.4%

4 Discussion

4.1 Accuracy Comparison

The accuracy scores indicate a high level of precision for the k-Nearest Neighbors classifier across different values of k. The results show that there is a consistency in the model's performance for k=1, 3, and 5, with a slight decrease when k is increased to 7.

4.2 Effect of Increasing k

Increasing the value of k from 1 to 7 shows a generally consistent performance, with a slight decrease in accuracy observed at k=7. This suggests that for the Digits dataset, a smaller neighborhood (lower k) provides slightly better or equivalent generalization compared to a larger neighborhood. However, the difference in performance is marginal, indicating robustness of the k-NN classifier to the choice of k in this specific context.

5 Conclusion

In conclusion, the k-Nearest Neighbors classifier demonstrates high accuracy across different values of k on the Digits dataset. The experiment reveals that the choice of k in k-NN has a minor impact on the model's accuracy in this case, with a slight decrease observed as k increases. These findings highlight the efficacy of k-NN for digit classification tasks and its relative insensitivity to the exact value of k within a certain range.